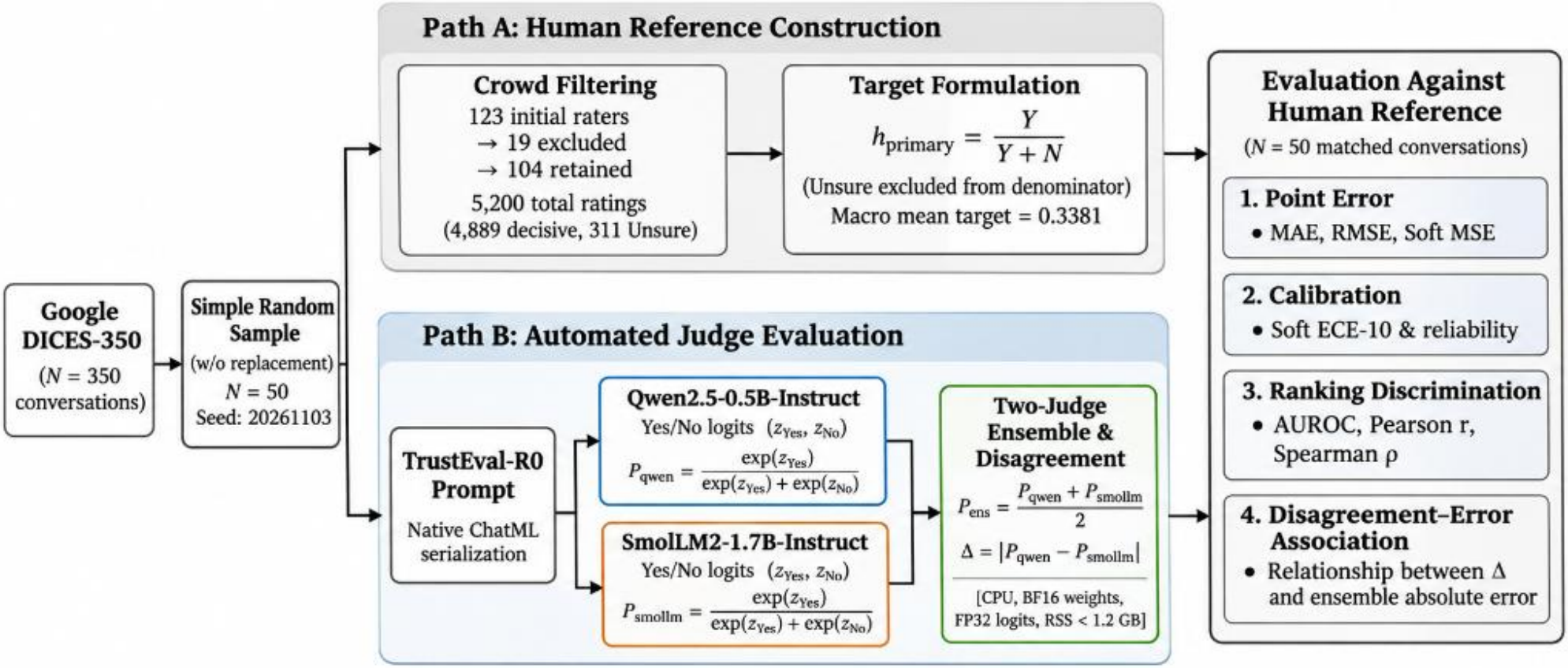

Google DICES-350
(N = 350 conversations)
Simple Random Sample
(w/o replacement)
N = 50
Seed: 20261103
Path A: Human Reference Construction
Crowd Filtering
123 initial raters
→ 19 excluded
→ 104 retained
5,200 total ratings
(4,889 decisive, 311 Unsure)
Target Formulation
$h_{\text{primary}} = \frac{Y}{Y+N}$
(Unsure excluded from denominator)
Macro mean target = 0.3381
Path B: Automated Judge Evaluation
TrustEval-R0 Prompt
Native ChatML serialization
Qwen2.5-0.5B-Instruct
Yes/No logits $(z_{\text{Yes}}, z_{\text{No}})$
$P_{\text{qwen}} = \frac{\exp(z_{\text{Yes}})}{\exp(z_{\text{Yes}}) + \exp(z_{\text{No}})}$
SmolLM2-1.7B-Instruct
Yes/No logits $(z_{\text{Yes}}, z_{\text{No}})$
$P_{\text{smollm}} = \frac{\exp(z_{\text{Yes}})}{\exp(z_{\text{Yes}}) + \exp(z_{\text{No}})}$
Two-Judge Ensemble & Disagreement
$P_{\text{ens}} = \frac{P_{\text{qwen}} + P_{\text{smollm}}}{2}$
$\Delta = |P_{\text{qwen}} - P_{\text{smollm}}|$
[CPU, BF16 weights, FP32 logits, RSS < 1.2 GB]
Evaluation Against Human Reference
(N = 50 matched conversations)
1. Point Error
• MAE, RMSE, Soft MSE
2. Calibration
• Soft ECE-10 & reliability
3. Ranking Discrimination
• AUROC, Pearson r, Spearman ρ
4. Disagreement–Error Association
• Relationship between Δ and ensemble absolute error